

Satellite over the pole

Y13 (2013) — English translation

A satellite with an orbital period of $T = 4$ hours is launched into orbit at the perigee point near the equator at low altitude. The earth's axis lies in the plane of this orbit.

A1 At what altitude h and after what time T_x after entering orbit can it fly over **0**
the pole for the first time?

The radius of the Earth is $R = 6400$ km, the orbital period of a satellite flying in a low circular orbit is $T_0 = 85$ minutes.

Source: <https://pho.rs/p/3996>